

PRACTICAL 1:

The **Books** table stores information about books such as book ID, title, ISBN, and publication year. The **Members** table stores member ID, name, and email.

To maintain **data integrity**, different constraints are applied:

- **PRIMARY KEY** uniquely identifies each book and member.
- **NOT NULL** ensures that every book has a title.
- **UNIQUE** prevents duplicate ISBN numbers and email addresses.
- **CHECK** ensures that the publication year is less than 2027.

Finally, the **INSERT** command is used to add sample records to the tables. These records help verify that the database and its constraints are working correctly.

```
mysql> -- Create the database
mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql>
mysql> -- Select the database
mysql> USE bookflow_db;
Database changed
mysql>
mysql> -- Create the Books table
mysql> CREATE TABLE Books (
  ->     book_id INT PRIMARY KEY,
  ->     title VARCHAR(255) NOT NULL,
  ->     isbn VARCHAR(20) UNIQUE,
  ->     published_year INT CHECK (published_year < 2027)
  -> );
ERROR 1050 (42S01): Table 'books' already exists
mysql>
mysql> -- Create the Members table
mysql> CREATE TABLE Members (
  ->     member_id INT PRIMARY KEY,
  ->     full_name VARCHAR(100),
  ->     email VARCHAR(100) UNIQUE
  -> );
ERROR 1050 (42S01): Table 'members' already exists
mysql>
```

```

mysql> -- Insert sample records into Books
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES
-> (1, 'Database Systems', '9780133970777', 2015),
-> (2, 'Learning SQL', '9780596520830', 2020),
-> (3, 'Introduction to Algorithms', '9780262046305', 2022);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
mysql>
mysql> -- Insert sample records into Members
mysql> INSERT INTO Members (member_id, full_name, email)
-> VALUES
-> (101, 'Alice Johnson', 'alice.johnson@example.com'),
-> (102, 'Brian Smith', 'brian.smith@example.com'),
-> (103, 'Catherine Lee', 'catherine.lee@example.com');
ERROR 1062 (23000): Duplicate entry '101' for key 'members.PRIMARY'
mysql>
mysql> -- Verify inserted data
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn                | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565      | 1925          |
| 2       | To Kill a Mockingbird | 9780061120084      | 1960          |
| 3       | 1984                 | 9780451524935      | 1949          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email                |
+-----+-----+-----+
| 101       | Alice Johnson  | alice@gmail.com      |
| 102       | Brian Smith    | brian@gmail.com      |
| 103       | Catherine Lee  | catherine@gmail.com  |
+-----+-----+-----+
3 rows in set (0.00 sec)

```